Demotion as Post-Redistribution Retirement, Not Primary Compression

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Abstract

The current lawful churn line in the Hilbert Substrate Framework (HSF) has reached a useful but incomplete stage. The minimal lawful move law now supports genuinely competing births and weakens, and it no longer rewards catastrophic core amputation or hidden hand-authored behavior. However, the current regime still exhibits accumulation with edge softening rather than full mesoscale churn: births occur, weakens occur, but node demotions do not become competitive. This note argues that the problem is not simply coefficient balance or insufficient demotion frequency. Rather, the present demotion move is conceptually miscast. It is being evaluated as an abrupt deletion of still-committed collective relief structure. Under the current witness stack, such a move should be expensive. The correct interpretation is that demotion is not a primary compression move. It is a *post-redistribution retirement* move, lawful only after a node's relief-carrying role has already been offloaded into surrounding structure. This note formalizes that stance, defines a demotion-readiness witness, and shows how demotion should be treated in future lawful implementations.

1 Status and Problem Statement

The current lawful churn program has stabilized enough to separate two different questions:

1. Is the move selector itself now lawful?
2. Given that selector, do the present witnesses support actual boiling/churn rather than monotone accumulation?

The answer to the first question is now provisionally yes. The frozen move set,

$$\{\text{no move},\ \text{birth one support patch},\ \text{weaken one interface},\ \text{demote one support patch}\},$$

is being evaluated under a common local counterfactual score law,

$$\Delta F(M) = \Delta E_{\text{expr}} - \lambda_B \Delta C_B - \lambda_S \Delta C_S - \lambda_F \Delta C_F - \lambda_R W_{\text{NR}}. \tag{1}$$

The recent lawful baseline is therefore no longer a hidden tuning script family, but a genuine implementation of a common move principle.

The answer to the second question is more limited. The present runs show:

- births can occur,
- weakens can occur,

- births and weakens can interleave,
- weakened edges can survive as reduced-strength interfaces,
- but node demotions do not become competitive.

Thus the current regime is best described as:

growth plus smooth edge compression, not yet full support-patch churn.

That is not a trivial failure. It is a highly informative one. It suggests that the current lawful line already contains a viable *edge-compression channel*, but does not yet contain a viable *node-retirement channel*.

2 The Core Diagnosis

The simplest incorrect reaction would be:

> "Demotions are too negative, so we should tune coefficients until they win sometimes."

That is not the right lesson.

The deeper issue is structural. In the present implementation, a demotion move is effectively scored as:

> remove the node, remove its incident support, remove its immediate interface participation, then evaluate the loss.

Under the present witness stack, that should be expensive whenever the node still participates in:

- collective expression,
- interface bookkeeping,
- organizer fidelity,
- relief-function support,
- or shell/core stability.

So the present strong negativity of demotion is not obviously a bug. In many cases it is the correct verdict on the wrong candidate move. The move being proposed is not:

> "retire a structure whose job has already been redistributed."

It is:

> "delete a structure that is still doing real work."

The lawful framework should reject that.

3 Demotion Is Not Primary Compression

We therefore adopt the following principle.

> **Demotion Principle.** Demotion is not a primary compression move. It is a post-redistribution retirement move.

This means:

- **Weakening** is the primary lawful compression channel.
- **Demotion** should become lawful only after weakening and surrounding re-expression have already evacuated the node's functional role.

In intuitive terms, a lawful demotion should not look like amputating a live limb. It should look like removing a scaffold that has already ceased to carry substantial relief.

This aligns naturally with the present no-refolding stance:

> committed collective relief structure is protected.

If a node still carries such committed relief structure, then a sudden deletion ought to be heavily penalized by both

$$\Delta C_F \quad \text{and} \quad W_{\mathrm{NR}}.$$

That is exactly what the present runs are telling us.

4 What the Current Regime Is Actually Doing

The current lawful baseline appears to realize the following sequence:

1. support patches are born when local expression gain outweighs burden and no-refolding penalties;
2. some edges become over-costly relative to their expression value;
3. those edges can be weakened rather than destroyed;
4. the graph thereby compresses locally without catastrophic function loss;
5. but no node becomes cheap enough to remove.

This is already meaningful. It shows that the framework can support:

- lawful growth,
- lawful soft compression,
- and protected persistence of still-committed structure.

What it does *not* yet show is full substrate boiling in the stronger sense of:

$$\text{birth, redistribution, retirement, rebirth.}$$

That stronger regime requires a lawful retirement condition.

5 The Missing Object: Demotion Readiness

The missing ingredient is therefore not "more demotion pressure" in the abstract. It is an explicit witness that a node has become *retireable*.

We introduce the following concept.

> **Demotion-readiness witness.** A scalar witness $W_{\mathrm{DR}}(v)$ associated to node/support patch v, measuring whether its relief-carrying role has already been sufficiently redistributed into surrounding structure.

A lawful demotion candidate should not be proposed merely because the node has low degree. Degree alone is too crude. A node can have degree 1 or 2 and still carry important organizer, shell, or bookkeeping function.

Instead, demotion readiness should be assessed functionally.

5.1 Required qualitative features

A node should become demotion-eligible only when all of the following are approximately true:

(a) **Interface attenuation:** its incident edges have already been weakened below a support threshold.

(b) **Low functional centrality:** its current contribution to local expression/organizer fidelity is small relative to nearby nodes.

(c) **Low bookkeeping dependence:** removing it does not substantially damage surrounding interface bookkeeping.

(d) **Low relief-function dependence:** its removal does not significantly degrade the local collective relief mode.

(e) **Local substitutability:** neighboring support can route or sustain the same functional role without requiring a discontinuous global rearrangement.

This is the correct notion of retirement. It says not "this node is weak," but rather:

> "this node's job has already been handed off."

6 Provisional Formalization

Let v denote a candidate node/support patch. We define a provisional demotion-readiness witness of the schematic form

$$W_{\mathrm{DR}}(v) = \alpha_I R_I(v) + \alpha_O R_O(v) + \alpha_B R_B(v) + \alpha_L R_L(v), \tag{2}$$

where:

- $R_I(v)$ measures **interface attenuation readiness**, e.g. whether incident edge strengths are already low;
- $R_O(v)$ measures **organizer/functional evacuation readiness**, e.g. whether organizer fidelity and local MI contribution have already drained away from v;

- $R_B(v)$ measures **bookkeeping safety**, e.g. whether removing v creates little additional interface-bookkeeping loss;
- $R_L(v)$ measures **relief-function substitutability**, e.g. whether the local relief mode is preserved by neighboring support after removing v.

The precise proxies remain open, but the structure matters:

demotion readiness must be built from already-redistributed function, not from bare sparsity.

6.1 Eligibility form

One clean way to implement this is:

$$\text{demotion of } v \text{ is lawful only if } W_{\mathrm{DR}}(v) \geq \theta_{\mathrm{DR}}, \tag{3}$$

for some provisional threshold θ_{DR}.

This means demotion does not simply compete with birth and weaken on raw score. It must first pass a retirement-readiness gate.

7 How Demotion Should Enter the Move Law

There are two principled ways to incorporate this.

7.1 Option A: Eligibility gate

The cleanest option is:

- only generate demotion candidates for nodes with $W_{\mathrm{DR}}(v) \geq \theta_{\mathrm{DR}}$;
- then evaluate those lawful candidates using the same move law (1).

This keeps the move law unchanged and interprets demotion-readiness as a precondition, not an extra reward term.

7.2 Option B: Readiness correction inside the score

Alternatively, define

$$\Delta F_{\mathrm{demote}}(v) = \Delta E_{\mathrm{expr}} - \lambda_B \Delta C_B - \lambda_S \Delta C_S - \lambda_F \Delta C_F - \lambda_R W_{\mathrm{NR}} + \lambda_D W_{\mathrm{DR}}(v), \tag{4}$$

with $\lambda_D > 0$.

This is less clean, because it makes demotion-readiness look like a bonus rather than a lawful precondition. For the present program, Option A is preferable.

8 Why Degree Alone Is Insufficient

The current implementation's demotion candidate restriction to low-degree nodes may be operationally useful, but it should not be regarded as the physical content of retirement.

The reason is simple:

- low degree does *not* imply low functional importance;
- low degree does *not* imply low bookkeeping cost of removal;
- low degree does *not* imply that surrounding support has already absorbed the node's role.

A node with degree 2 can still be:

- a bridge in the local organizer pattern,
- part of the dominant shell,
- a nontrivial carrier of mutual information,
- or a crucial participant in a relief-carrying triangle.

Thus the relevant question is not:

"How many edges does the node have left?"

but:

"Has the node already become functionally redundant?"

That is a fundamentally different criterion.

9 Implications for the Boiling Picture

If this note is correct, then the stronger boiling picture should be refined as follows.

The substrate does not churn by directly birthing and deleting support patches in a symmetric way. Instead it churns in a layered sequence:

1. **Birth:** a new support patch becomes favorable because it increases local expression and/or relieves burden.
2. **Redistribution:** surrounding edges and local support re-express through weakens and altered organizer structure.
3. **Evacuation:** some older support patch loses relief-carrying relevance as surrounding structure absorbs its function.
4. **Retirement:** only then does demotion become cheap and lawful.

This is a much more believable mesoscopic churn mechanism. It is also closer to the present status-document requirement that support should undergo ongoing re-expression rather than abrupt arbitrary rewrites.

In short:

boiling is not birth versus deletion;
it is birth, redistribution, and only then retirement.

10 Research Consequences

This has immediate implications for the next lawful implementation.

10.1 What should not be done

We should *not*:

- retune coefficients merely to make demotion occur more often;
- relax no-refolding penalties so that abrupt node deletion becomes easy;
- interpret the absence of demotion as evidence that the lawful selector is broken.

Any of those would risk reintroducing exactly the kind of hand-authored behavior that the derivation-first program was meant to eliminate.

10.2 What should be done

We should:

- preserve the current lawful baseline move law;
- preserve weaken as the primary lawful compression channel;
- define a provisional demotion-readiness witness W_{DR};
- allow demotion only after local redistribution has already evacuated the node's role.

That is the narrow, honest next step.

11 Provisional Interface to Future Scripts

A future lawful script can implement this note with minimal disturbance to the v4 baseline:

1. Keep the frozen move set unchanged.
2. Keep the same score law for births and weakens.
3. Replace raw demotion candidacy by:

   $$\texttt{candidate\_demotions} \to \texttt{candidate\_retirements}$$

   based on a demotion-readiness witness.
4. Permit only nodes satisfying

   $$W_{\mathrm{DR}}(v) \geq \theta_{\mathrm{DR}}$$

   to enter the demotion comparison set.
5. Continue logging raw and witness-resolved reasons for acceptance/rejection.

This would allow the next script revision to remain tightly scoped:

not new physics, not a new move law, just a lawful retirement gate.

12 Conclusion

The present lawful churn line has already achieved something important: it supports births and smooth weakens under a common move law, without hidden selectors and without rewarding core amputation.

Its present limitation is equally informative. The absence of node demotion does not necessarily mean the compressive side of the dynamics is missing. Rather, it indicates that compression is currently implemented only at the level of edges, not yet at the level of retired support patches.

That suggests a clean conceptual correction:

> **Demotion should not be interpreted as primary compression.**
> **It should be interpreted as post-redistribution retirement.**

Once phrased that way, the path forward is much clearer. The next lawful implementation should not force demotion by tuning. It should introduce a demotion-readiness witness that recognizes when a node's relief-carrying role has already been redistributed, and only then allow retirement to compete.

That is the right route from edge-softening toward genuine mesoscale churn.